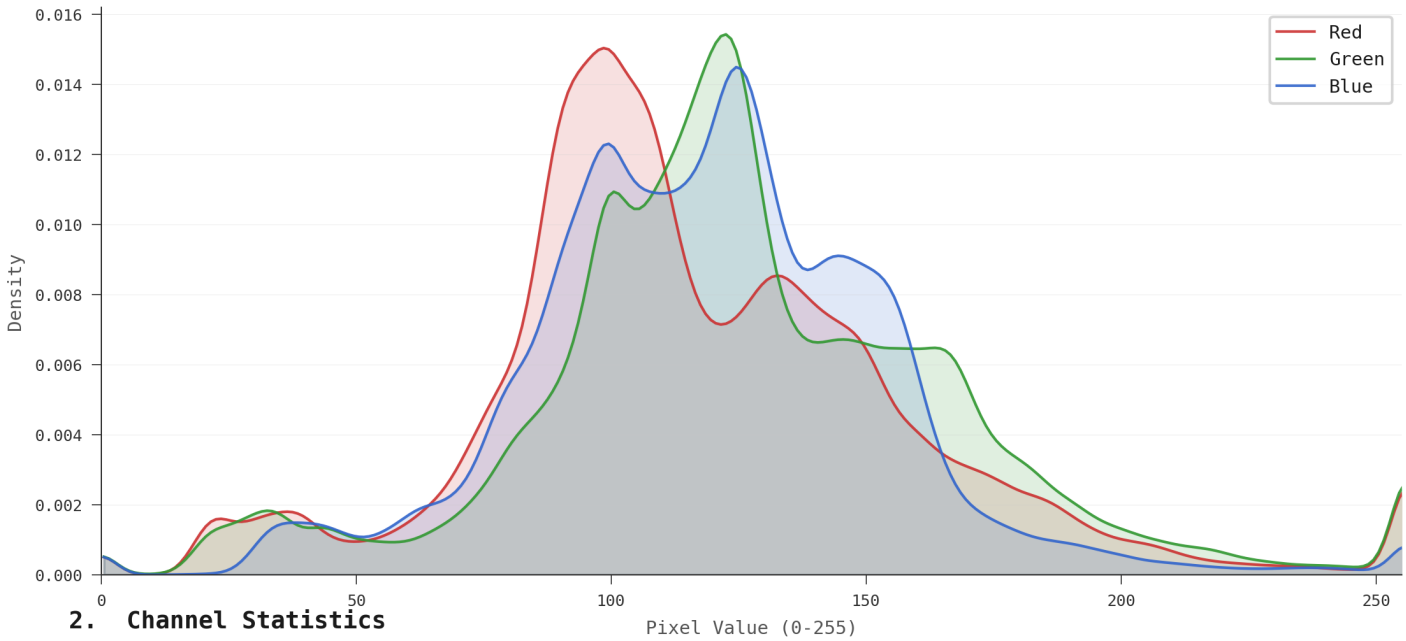
Reference Data Sheet: KODIM10

Subject: Sailboat Race Close-Up

512x768 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	115.6475	40.3658	1629.3985	1.1267	0.4697	0	255
Green	124.7658	40.9145	1673.9978	0.9920	0.1665	0	255
Blue	118.2671	33.8910	1148.6022	1.3551	0.2009	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9555	0.8902
G	0.9555	1.0000	0.9695
B	0.8902	0.9695	1.0000

Pairwise Pearson Correlation Coefficients

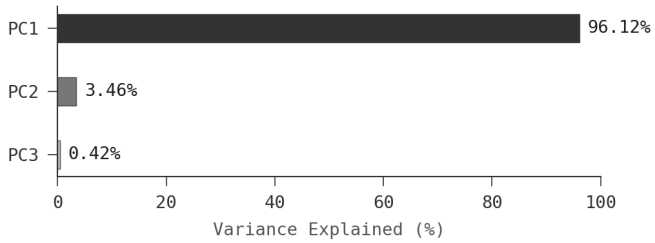
r(R-G)	=	**0.955452**
r(R-B)	=	**0.890182**
r(G-B)	=	**0.969513**
Avg |r|	=	**0.938382**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1629.4026} & 1577.9784 & 1217.8063 \\
1577.9784 & \mathbf{1674.0021} & 1344.3638 \\
1217.8063 & 1344.3638 & \mathbf{1148.6052}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM10

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6010	**-0.6226**	-0.5011	4279.3978
PC2	-0.7275	0.1666	0.6656	153.9494
PC3	0.3309	-0.7646	0.5530	18.6628

6. Derived Metrics

Condition Number

229.30

(l1/l3) PC1/PC2: 27.80 | PC2/PC3: 8.25

Blue Channel Independence

6.4%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6226 | Green dominant

Classification: Strongly One-Dimensional (PC1 = 96.12%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.953891	0.965941	0.959916
Green	0.953087	0.963732	0.958410
Blue	0.935530	0.951529	0.943529

8. Average Local Variance (3x3)

Channel	Value
Red	107.2469
Green	116.0858
Blue	103.4320

9. Redundancy Profile

Classification: Strongly One-Dimensional | Condition Number: 229.30 | PC1: 96.12%

Correlation: G–B pair dominates at r = 0.969513 | R–G: 0.955452 | R–B: 0.890182

Blue Channel: 6.4% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.953952 | Avg local variance: 108.9216

Eigenvector: Green dominant | G (-0.6226) leads | R (-0.6010), B (-0.5011)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."